LECTURE 3

Work and Energy

❖ Work



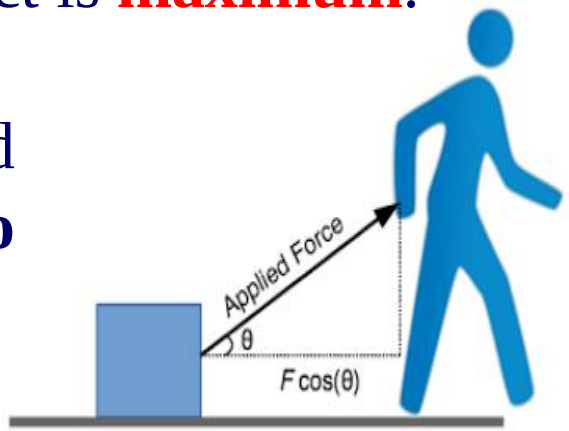
- It is the energy transferred to or from an object via the application of force along a displacement.
- It is the product of the magnitude **F** of the force, the magnitude **Δr** of the displacement of the point of application of the force, and **cos θ**, where **θ** is the angle between the force and displacement vectors.
- Work is a scalar quantity

$$W \equiv F\,\Delta r \cos\theta$$

- the SI unit of work is the newton. meter (N· m) ≈ joule (J).

- The work done by a force on a moving object is **zero** when the force applied is perpendicular to the displacement of its point of application.
- If an applied force **F** is in the same direction as the displacement **Δr**, then **θ = 0** and cos 0 = 1. In this case,

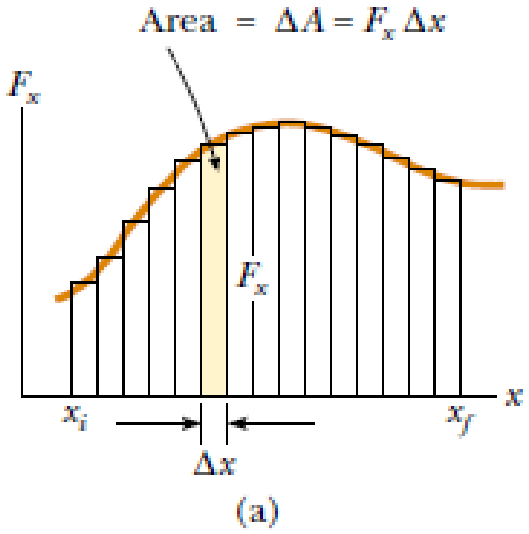
the work done by a force on a moving object is **maximum**.



- If **W** is the work done on a system and W is **positive**, energy is transferred **to** the system.
- If **W** is **negative**, energy is transferred **from** the system.

Work Done by a Varying Force

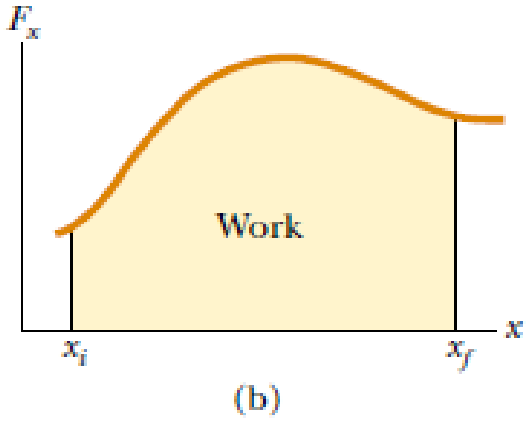
- Consider a particle being displaced along the x axis under the action of a force that varies with position. The particle is displaced from $x = x_i$ to $x = x_f$.



(a)

- the work done by F_x as the particle moves from x_i to x_f is given by :

$$W = \int_{x_i}^{x_f} F_x \, dx$$



(b)

- The work done by the component F_x of the varying force as the particle moves from x_i to x_f is exactly equal to the area under this curve.

Work Done by a Spring

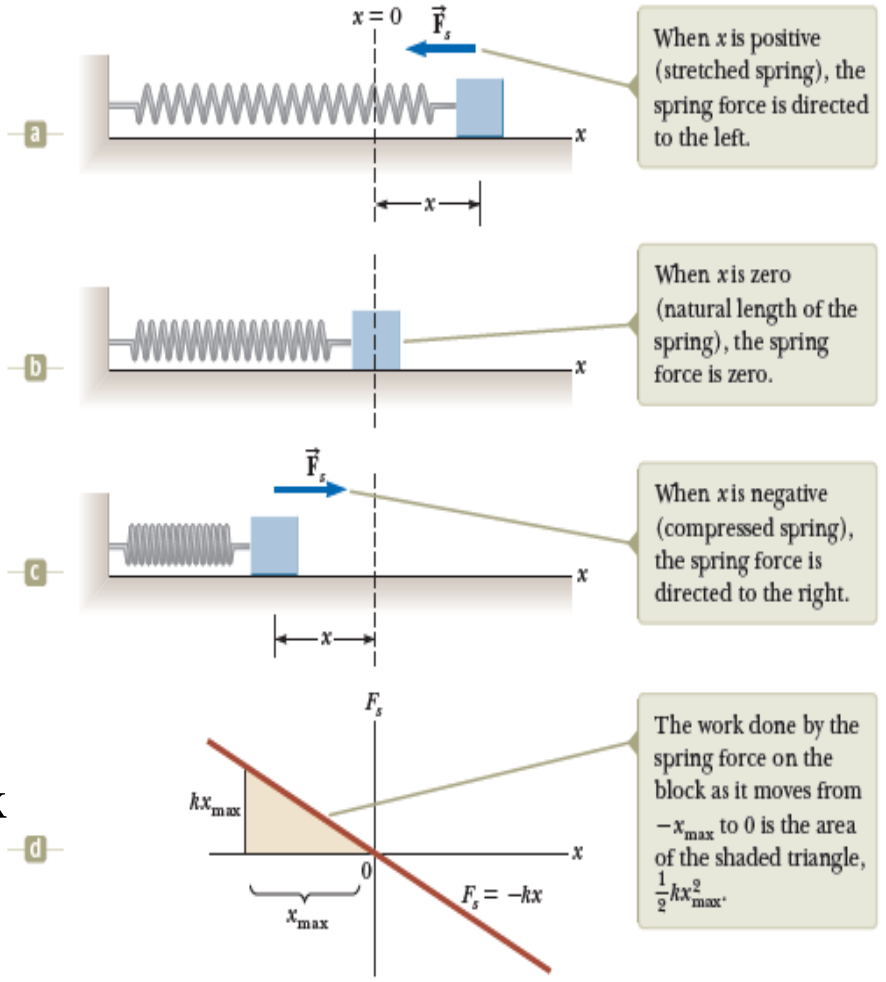
The force required to stretch or compress a spring is proportional to the amount of stretch or compression *x*. This force law for springs is known as

Hooke's law.

$$F_s = -kx$$

If the block undergoes an arbitrary displacement from x = x_i to x = x_f , the work done by the spring force on the block is

$$W_{ext} = \int_{x_i}^{x_f} kx\,dx = \tfrac{1}{2}kx_f^2 - \tfrac{1}{2}kx_i^2$$

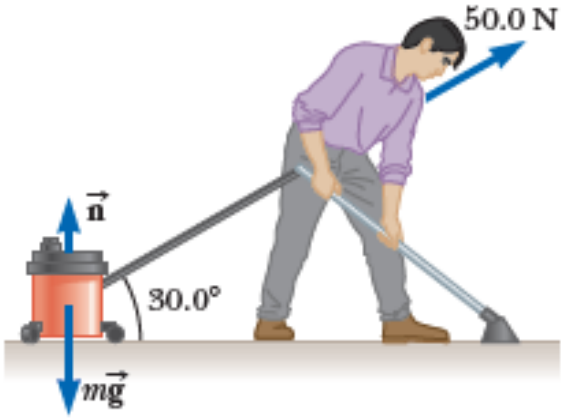


Example

A man cleaning a floor pulls a vacuum cleaner with a force of magnitude $F = 50.0$ N at an angle of 30.0^{o} with the horizontal. Calculate the work done by the force on the vacuum cleaner as the vacuum cleaner is displaced 3.00 m to the right.

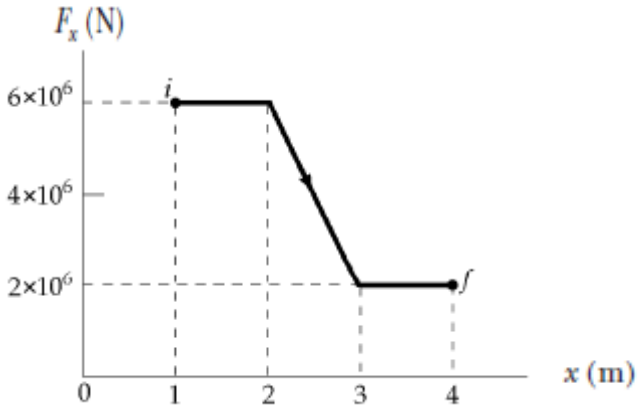
Sol.

$$W = F\,\Delta r \cos\theta = (50.0\ \text{N})(3.00\ \text{m})(\cos 30.0^{\circ})$$
$$= 130\ \text{J}$$



Example

particle is subject to a force F_x that varies with position as shown in Figure Find the total work done by the force on the particle as it moves from initial point to final point.

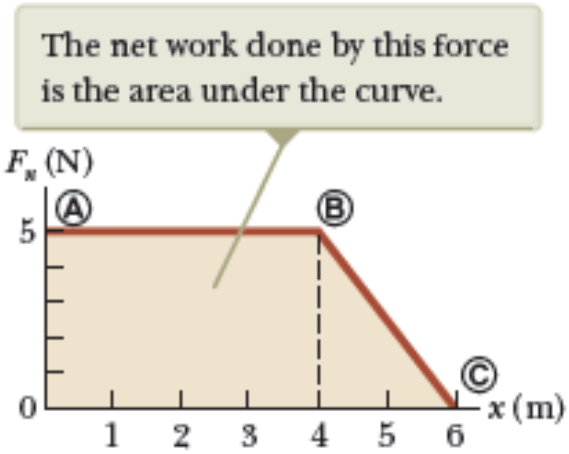


Sol.

$$W_{\text{Ⓐ to Ⓑ}} = (5.0\ \text{N})(4.0\ \text{m}) = 20\ \text{J}$$

$$W_{\text{Ⓑ to Ⓒ}} = \tfrac{1}{2}(5.0\ \text{N})(2.0\ \text{m}) = 5.0\ \text{J}$$

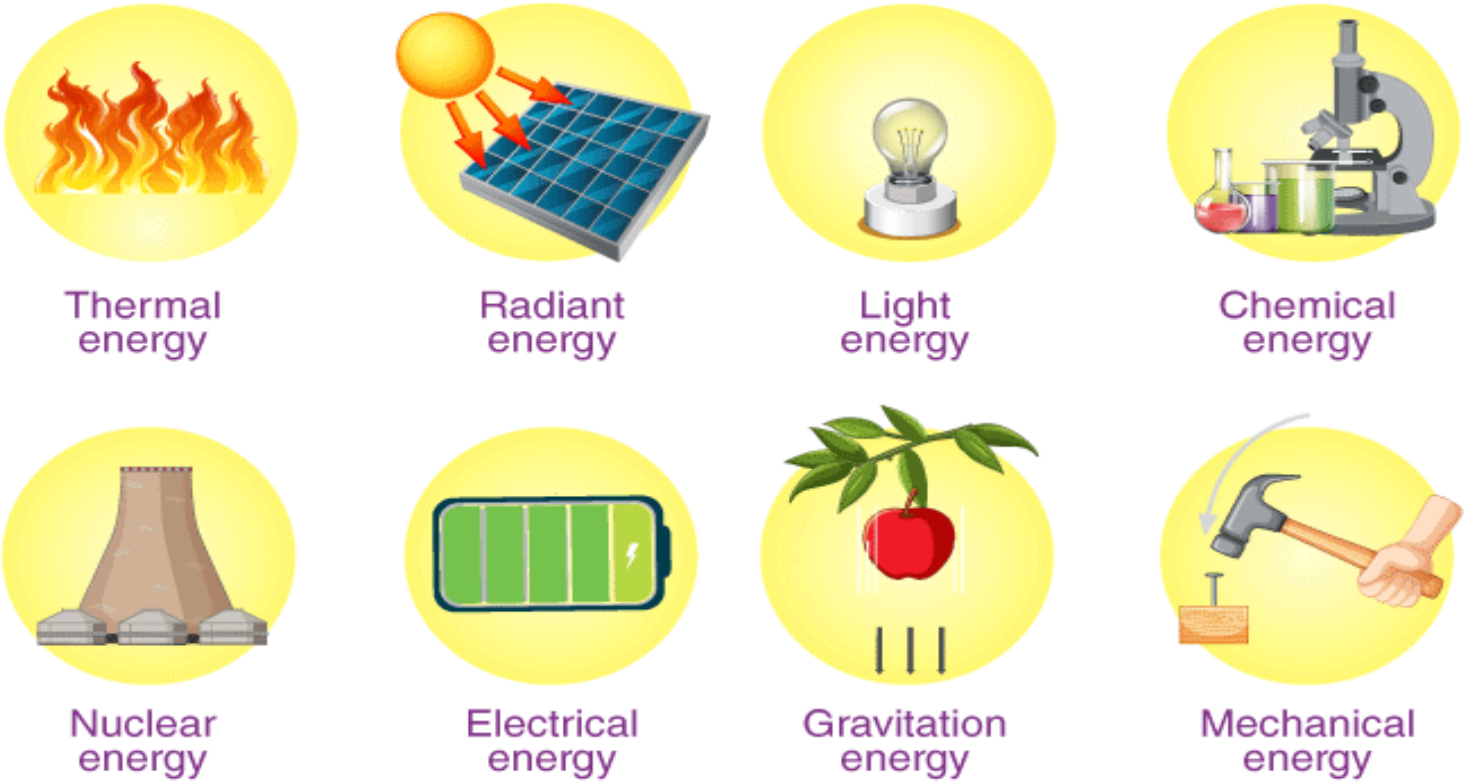
$$W_{\text{Ⓐ to Ⓒ}} = W_{\text{Ⓐ to Ⓑ}} + W_{\text{Ⓑ to Ⓒ}} = 20\ \text{J} + 5.0\ \text{J} = 25\ \text{J}$$



❖ Energy

Energy is the ability to do work.





These forms of energy can be grouped into two general types of energy for doing work:

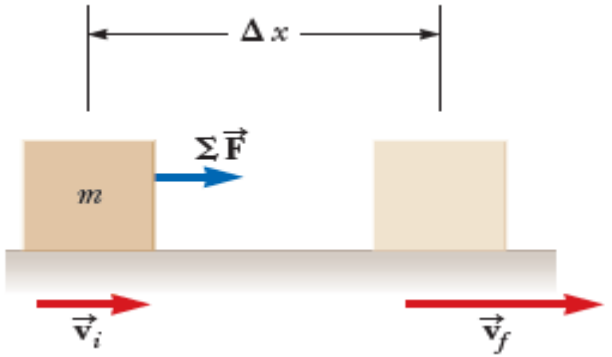
Kinetic, or working energy

Potential, or stored energy

Kinetic energy

- Kinetic energy represents the energy associated with the motion of the particle
- kinetic energy is a scalar quantity and has the same units as work

$$W_{\text{ext}} = \int_{x_i}^{x_f} \sum F \, dx$$



Using Newton's second law, we substitute for the magnitude of the net force $\Sigma F = ma$ and then perform the following chain-rule manipulations on the integrand:

$$W_{\text{ext}} = \int_{x_i}^{x_f} ma \, dx = \int_{x_i}^{x_f} m \frac{dv}{dt} dx = \int_{x_i}^{x_f} m \frac{dv}{dx} \frac{dx}{dt} dx = \int_{v_i}^{v_f} mv \, dv$$

$$W_{\text{ext}} = \tfrac{1}{2}mv_f^2 - \tfrac{1}{2}mv_i^2$$

- the work done on a particle by a net force Σ F acting on it equals the change in kinetic energy of the particle.

$$W_{\text{ext}} = K_f - K_i = \Delta K$$

Kinetic energy ▶ $K = \frac{1}{2}mv^2$

work–kinetic energy theorem

When work is done on a system and the only change in the system is in its speed, the net work done on the system equals the change in kinetic energy of the system

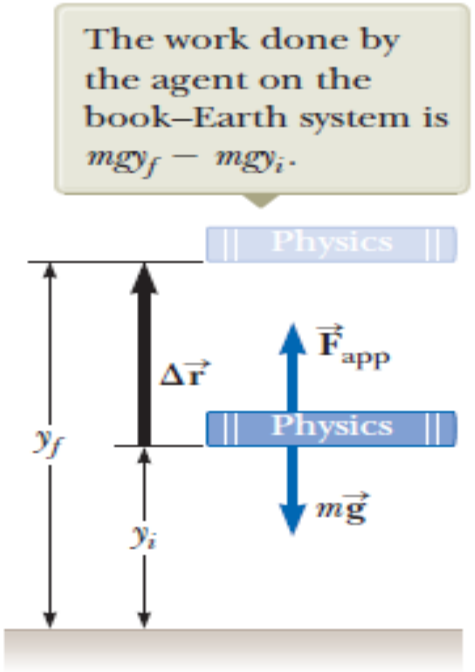
$$W_{\text{ext}} = K_f - K_i = \Delta K$$

The work–kinetic energy theorem indicates that:

- the speed of a system *increases* if the net work done on it is *positive* because the final kinetic energy is greater than the initial kinetic energy.
- The speed decreases if the net work is negative because the final kinetic energy is less than the initial kinetic energy.

Potential energy

Potential energy reflects the potential of an object to have motion, and generally is a function of the position of an object within a field or may be stored in the field itself.



$$W_{ext} = (\vec{F}_{app}) \cdot \Delta\vec{r} = (mg\hat{j}) \cdot [(y_f - y_i)\hat{j}] = mgy_f - mgy_i$$

Gravitational potential energy ▶ $U_g \equiv mgy$

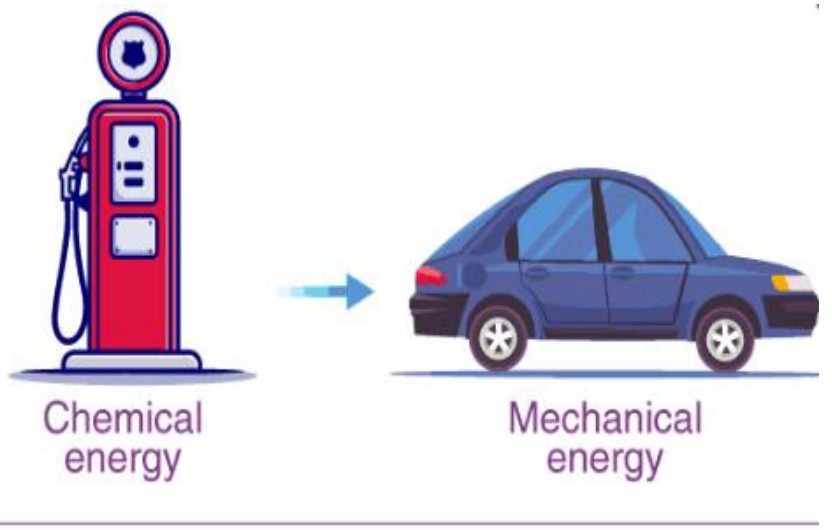
Conservation law of Energy

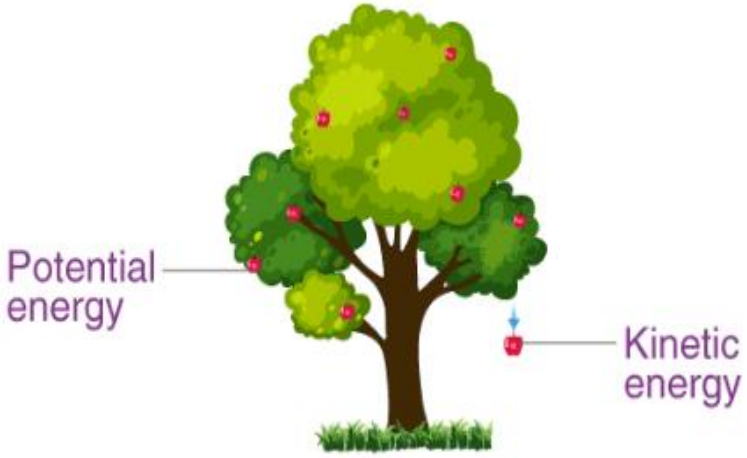
The law of conservation of energy states that:

energy can not created or destroyed.

it may be transformed from one form to another.

If you take all forms of energy into account, the total energy of an isolated system remains constant; it is said to be conserved over time

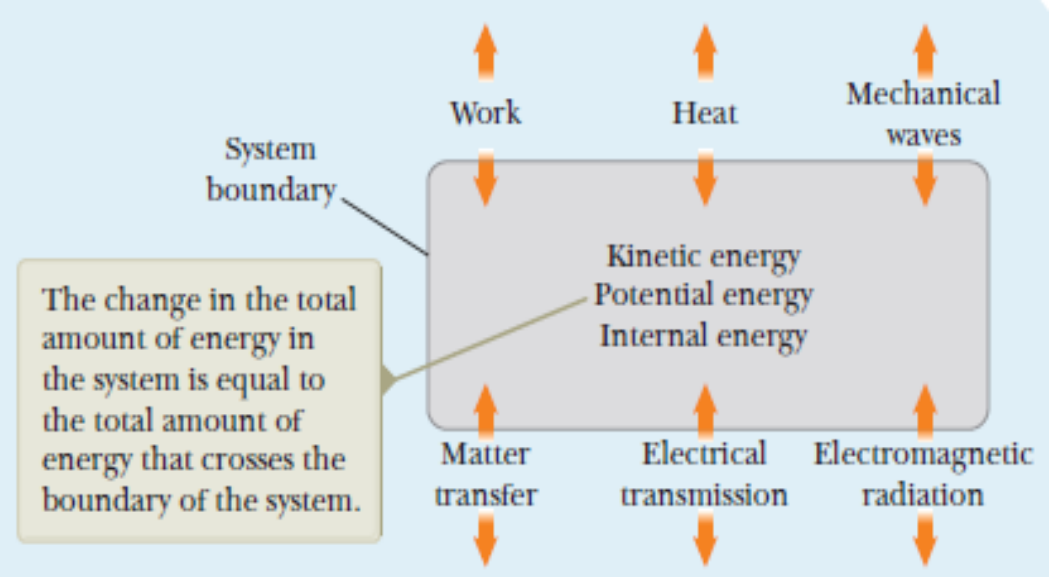




Nonisolated System

The most general statement describing the behavior of a nonisolated system is the **conservation of energy equation**

$$\Delta E_{\text{system}} = \Sigma T$$



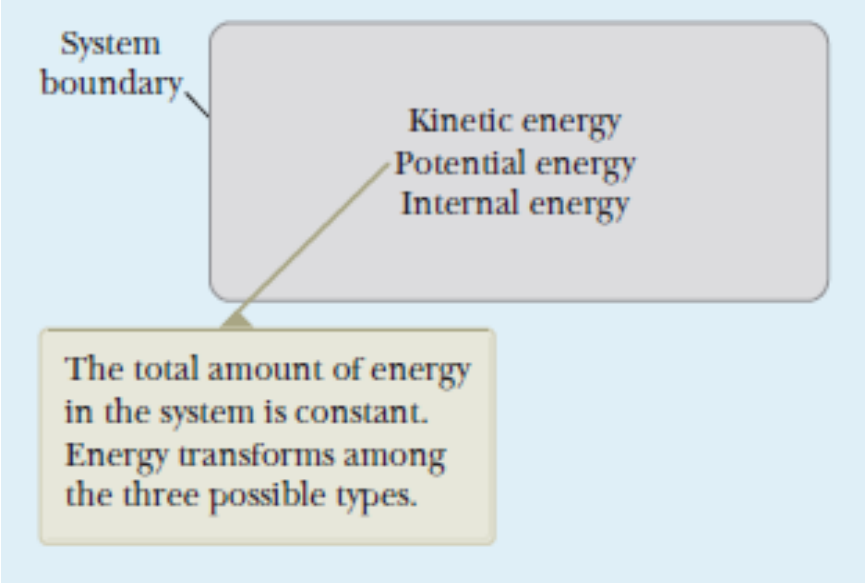
Including the types of energy storage and energy transfer

$$\Delta K + \Delta U + \left|\Delta E_{\text{int}}\right. = W + Q + T_{\text{MW}} + T_{\text{MT}} + T_{\text{ET}} + T_{\text{ER}}$$

Isolated System

The total energy of an isolated system is conserved, so

$$\Delta E_{\text{system}} = 0$$



which can be written as

$$\Delta K + \Delta U + \Delta E_{\text{int}} = 0$$


Thank
You